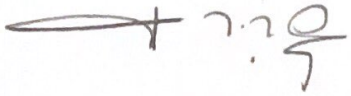
" On my honor, I have neither given nor received any unauthorized and/or inappropriate assistance for all sessions of this exam. The work done on this exam is totally my own, I understand that by the school code, violation of these principles will lead to a zero grade and is a subject to harsh discipline issues."

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Q1] $I_1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -8 & 1 & 0 & 0 \\ -4 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix}$, $I_2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1/2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

$$I_3 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -1/2 & 1 \end{bmatrix}$$

$I_3 . I_2 . I_1 . A = U$

Q2]

a) $x = 5y \Rightarrow x - 5y = 0$

$x = 5y$

General solution $\Rightarrow \left\{ y \begin{pmatrix} 5 \\ 1 \end{pmatrix} \right\}$ → basis

Minimum dimension is $\underline{\underline{1}}$ $x = 0$ and $y = 0$

c) $x + 4y - z = 6$

since $x = y = z \Rightarrow$ can not generate 0 (zero) vector in $\mathbb{R}^3$, the system DOESN'T FORM A VECTOR SPACE

(*) $x = 6 - 4y + z$, $y = y$, $z = z$

$$\bar{X} = \begin{bmatrix} 6 - 4y + z \\ y \\ z \end{bmatrix}$$ 1 – 1 dimension

Q3] u is a unit vector in $\mathbb{R}^n$

$H = I - 2uu^T$ ($n \times n$ symetric matrix)

a) $H^2 = H \cdot H = (I - 2uu^T)(I - 2uu^T)$

$= I^2 - I\,2uu^T - 2uu^T I + 4uu^T uu^T$

$= I - 2uu^T - 2uu^T + 4uu^T uu^T$

$= I - 4uu^T + 4u\underbrace{u^T u}_{1}u^T$ since u is unit vector

$= I - 4uu^T + 4uu^T$ so $u^T u = \|u\|^2 = 1$

$H^2 = I$

b) By the definition of the orthogonal matrix $A^TA = I$

Depending on this, $H^TH = I$ so

$(I - 2uu^T)^T(I - 2uu^T) = (I^T - 2uu^T)(I - 2uu^T)$

$= (I - 2uu^T)(I - 2uu^T) = I^2 - I\,2uu^T - 2uu^T I - 2uu^T - 2uu^T$

$= I - 2uu^T - 2uu^T + 4u\underbrace{u^T u}_{1}u^T$

$= I - 4uu^T + 4uu^T = I$, thus $H^TH = I$

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c) One eigenvector of H is "u"

If u is the eigenvector of H, then $Hu = \lambda u$, here λ is the corresponding eigenvalue. $H = I - 2uu^T$

$$(I - 2uu^T)u = \lambda u \Rightarrow (I - 2uu^T)u - \lambda u = 0$$

$$Iu - 2uu^T u - \lambda u = 0$$

$$u - 2u\underbrace{u^T u}_{1} - \lambda u = 0 \quad \text{since } u \text{ is unit vector } \|u\| = 1$$

$$u^T u = \|u\|^2 = 1$$

$$u - 2u - \lambda u = 0 \Rightarrow -u - \lambda u = 0 \Rightarrow (-1 - \lambda)u = 0$$

since u is unit vector, to hold the equality so $-1 - \lambda = 0 \Rightarrow \boxed{\lambda = -1}$

Therefore corresponding eigenvalue is $\lambda = -1$.

d) Yes, it is diagonalizable since H matrix is both symmetric and orthonormal matrix. A symmetric matrix $n \times n$ has exactly n eigenvalues. And also, it is orthogonally diagonalizable if and only if it is symmetric.

Q4]

a) first tank $v(0) = 10$

Second tank $w(0) = 40$

$$\frac{d}{dt}\begin{bmatrix} v \\ w \end{bmatrix} = \underbrace{\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}}_{A}\begin{bmatrix} v \\ w \end{bmatrix}$$ this is the process

b, c)

$$\det(A - \lambda I) = \begin{vmatrix} -1-\lambda & 1 \\ 1 & -1-\lambda \end{vmatrix} = (1+\lambda)^2 - 1 = 0$$

$$\lambda(\lambda+2) = 0, \quad \lambda_1 = 0 \quad \lambda_2 = -2$$

for $\lambda_1 = 0$ $\quad v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

for $\lambda_2 = -2$ $\quad v_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$